llmXive Automated Review of Orca: The World is in Your Mind

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (8 lenses): Claim Accuracy, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses strictly on the accuracy of factual claims and the validity of citations.

Fatal Citation and Factuality Issues: The most critical issue is the reliance on citations for models and benchmarks that appear to be hallucinated or from the future (2025-2026). The bibliography includes entries for `GPT54` (OpenAI, March 2026), `Gemini-3.1-pro` (Google, November 2025), `Qwen3.5` (Feb 2026), and `DeepSeek-V4` (2026). As of the current date, these models do not exist in the public domain. The paper presents quantitative comparisons (e.g., Table 1, Table 3) against these non-existent baselines. For instance, claiming Orca outperforms "GPT 5.4" on the PRICE-V0.1 benchmark is factually impossible to verify and suggests the data is synthetic or hallucinated. This undermines the entire empirical validity of the paper.

Internal Logical Inconsistencies: In Section 5.1.1 (Text Generation), the text claims Orca achieves the "best overall result among same-size VLMs and large-size world models." However, Table 1 shows `V-JEPA 2.1` scoring 75.4 on MVBench, while `Orca-4B` scores 65.3. While the text attempts to qualify this by saying "among same-size," the inclusion of V-JEPA 2.1 (10B) in the same table without a clear distinction in the narrative creates confusion. The claim that Orca is "best" is not supported by the raw numbers presented in the table for the MVBench metric, where a baseline outperforms the proposed model significantly.

Missing Citations: The citation key `judge1.5` is referenced in Section 5.1 (Metrics) and Table 4 ("PRM-as-a-Judge series [PRM-as-a-Judge, judge1.5]") but is absent from the provided `TR_Ref.bib` file. This makes the claim regarding the specific diagnostic metrics unverifiable.

Conclusion: The paper makes strong empirical claims based on comparisons with models that do not exist and contains internal contradictions between its textual conclusions and the provided data tables. The science cannot be evaluated as sound until the existence of the baselines and the accuracy of the reported numbers are verified.

Action items.

- [fatal] Citation keys for future-dated or non-existent models (e.g., 'GPT54', 'Gemini-3.1-pro', 'Qwen3.5', 'DeepSeek-V4') are unsupported by the provided bibliography. The bib file contains entries for these, but they cite 2025/2026 dates and URLs that do not exist in the public domain as of the current date. Claims of performance against these baselines (e.g., Table 1, Table 3) are factually unverifiable and likely hallucinated.
- [science] The claim that Orca-4B achieves a 51.8 average score on text benchmarks (Table 1) and outperforms 'V-JEPA 2.1' (75.4 on MVBench) is internally inconsistent. The text states Orca is 'best among same-size VLMs and large-size world models,' yet V-JEPA 2.1's MVBench score (75.4) is significantly higher than Orca's (65.3). The comparison logic in the text does not support the conclusion drawn.
- [writing] The citation 'judge1.5' is used in Section 5.1 (Metrics) and Table 4 but is missing from the provided bibliography (TR_Ref.bib). The claim that PRM-as-a-Judge and 'judge1.5' are used for diagnostics cannot be verified.
- [science] The paper claims 'no action-labeled data was used in pre-training' (Section 5.1.2) yet cites 'pi0.5' (a VLA model trained on robot data) as a baseline. While this is a valid comparison, the text implies Orca's action capabilities emerge solely from video,

but the baseline comparison does not isolate the effect of action-label pre-training in the baselines (e.g., V-JEPA 2.1 vs Orca) clearly enough to support the 'emergence' claim without further ablation.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a high density of specialized acronyms and domain-specific jargon that are not defined upon their first appearance, creating a barrier for non-specialist readers. While the technical content is advanced, the presentation assumes a level of familiarity that excludes broader audiences.

Specific instances requiring attention include:

1. Undefined Acronyms: Terms like "OOD" (out-of-distribution), "VLA" (Vision-Language-Action), "MBA" (Multiple Binary Accuracy), "TI2I" (instruction-conditional image-to-image), "LoRA" (Low-Rank Adaptation), "PRM" (Process Reward Model), "FSDP" (Fully Sharded Data Parallel), "DiT" (Diffusion Transformer), and "MMDiT" appear frequently without definition. For example, in Section 3.2, "OOD Settings" is introduced without explanation. In Section 4.2.1, "TI2I" is used in the benchmark description without context.
2. Jargon Overuse: The term "readout" is used repeatedly (e.g., Section 4.2, 4.3) to describe the mechanism for generating outputs. While standard in some contexts, replacing it with "decoder," "interface," or "generation head" in specific instances would improve clarity for a general audience.
3. Ambiguous Phrasing: The phrase "state transition" is used extensively. While technically precise, varying the language with terms like "state evolution" or "dynamic change" could enhance readability without losing meaning.

Addressing these issues by defining acronyms at first use and simplifying repetitive jargon will significantly improve the paper's accessibility without compromising its scientific rigor.

Action items.

- [writing] Define 'OOD' (out-of-distribution) at first use in Section 3.2 (Appendix Evaluation Settings) and Section 4.2.3.
- [writing] Define 'VLA' (Vision-Language-Action) at first use in Section 3.2 and Section 4.2.3.
- [writing] Define 'MBA' (Multiple Binary Accuracy) at first use in Section 3.2 under Tem-

poralBench description.

- [writing] Define 'TI2I' (instruction-conditional image-to-image) at first use in Section 4.2.1 (PRICE-V0.1 description).
- [writing] Define 'LoRA' (Low-Rank Adaptation) at first use in Section 4.2.2 (Vision Readout settings).
- [writing] Define 'PRM' (Process Reward Model) at first use in Section 3.2 (Metrics) and Section 4.2.3.
- [writing] Define 'FSDP' (Fully Sharded Data Parallel) at first use in Section 4.1 (Infrastructure).
- [writing] Define 'DiT' (Diffusion Transformer) at first use in Section 4.2.3 (Action Readout).
- [writing] Define 'MMDiT' at first use in Section 4.2.2 (Vision Readout settings).
- [writing] Replace 'readout' with 'decoder' or 'interface' in several instances (e.g., Section 4.2, 4.3) to reduce jargon density for non-specialists.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

"The paper presents a coherent logical flow from the 'world latent' hypothesis to the multimodal readout results. The central claim—that pre-training on state transitions (without action labels) yields a latent space that improves downstream text, image, and action tasks—is supported by the scaling laws (Figures 1-2) and the ablation study. The logic that 'stronger latent leads to stronger readouts' (Answer 1.2) follows directly from the observed correlation between pre-training loss and downstream performance.\n\nHowever, there are minor logical inconsistencies in the presentation of comparative results and the interpretation of ablation data:\n\n1. Comparative Claims vs. Data: In Section 3.3 (Action Generation), the text states Orca 'outperforms Qwen3.5 in all OOD settings,' which is true. However, the subsequent sentence says it is 'comparable to pi_0.5.' In the Object OOD setting, Orca (28.2) actually underperforms pi_0.5 (31.2) on the primary Rule-based metric, though it wins on DRR and SQS. The phrasing 'outperforms... in all OOD settings' could be misread as 'outperforms all baselines in all settings.' The logic holds that Orca is competitive, but the text should be precise: 'outperforms VLM baselines and approaches the performance of the specialized VLA baseline pi_0.5.'\n\n2. Ablation Interpretation: Table 4 shows that removing lambda_vqa results in a failure ('-') for Image Prediction. The text concludes that VQA 'strengthens semantic grounding.' While likely true, the logical gap is

that the failure might be architectural (e.g., the image readout head requires VQA tokens as input) rather than purely representational. The paper should clarify if the image readout is structurally dependent on the VQA stream or if the latent representation simply collapses without it. If the latter, the claim is strong; if the former, the 'grounding' argument is less direct.\n\n3. Throughput Attribution: The claim of a 4.4x speedup over StarVLA (Section: Infrastructure) compares the 'Full Orca' stack against StarVLA. If StarVLA does not use FSDP2 or the same underlying infrastructure optimizations, the comparison conflates the proposed optimizations with the base framework. The logic requires that the baseline be isolated to the specific optimizations claimed (e.g., '4.4x over StarVLA using the same FSDP2 baseline').\n\nOverall, the causal mechanisms proposed (state transition pre-training -> better latent -> better readouts) are logically sound and well-supported by the provided data, provided the comparative phrasing is tightened."
Action items.

- [writing] "The paper presents a coherent logical flow from the 'world latent' hypothesis to the multi-modal readout results. The central claim—that pre-training on state transitions (without action labels) yields a latent space that improves downstream text, image, and action tasks—is supported by the scaling laws (Figures 1-2) and the ablation study. The logic that 'stronger latent leads to stronger readouts' (Answer 1.2) follows directly from the observed correlation between pre-training loss and downst

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several significant over-claims regarding the causal efficacy of its proposed "world latent" and the generalization capabilities of the Orca model.

First, the central claim that the model learns "without action labels during pre-training" and that this specific mechanism is responsible for its superior real-robot performance (Table 4, Sec:evaluation_toA) is not fully supported. While the pre-training is indeed action-free, the evaluation of action generation relies on an "Action Expert" trained on 200 real-robot trajectories per task. The paper attributes the performance gain (e.g., 36.6 vs 27.6 rule-based score) to the "world latent," but fails to isolate this variable. It is equally plausible that the improvement stems from the specific initialization of the latent space or the architecture of the readout head, rather than the "world learning" paradigm itself. The claim that the paradigm "alleviates low generalization due to robot data scarcity" (Sec:evaluation_scaling_performance) is an overreach; the model still requires significant downstream fine-tuning data (200 trajectories) to function, and the paper does not demonstrate that it requires *less* data than baselines to achieve similar performance.

Second, the scaling analysis (Answer 1.2) claims that "stronger world latent... leads to stronger downstream readouts.” The evidence provided compares Orca against baselines that use the *same* Action Expert architecture and training data. Without an ablation study showing that a random initialization or a non-world-model latent (e.g., standard VLM) performs worse *given the same readout training*, the claim that the “world latent” is the primary driver of success is speculative. The paper over-attributes the results to the pre-training objective without ruling out confounding factors in the readout training process.

Finally, the “Future Works” section (Sec:conclusion) lists “Quantum” and “Proteins” as potential domains for the model’s readout. Given that the model is a 4B parameter system trained on video and text, this is a gross overreach. There is no theoretical or empirical basis provided to suggest that a latent space learned from macroscopic video dynamics can represent quantum states or protein folding. This claim detracts from the scientific rigor of the paper and should be removed or significantly qualified as purely speculative.
Action items.

- [science] Claiming the ‘world latent’ alone drives robot success is overreach. The Action Expert uses 200 trajectories/task. The gain may stem from fine-tuning data or architecture, not just pre-training. Clarify the causal link or reduce the claim.
- [science] The paper states the model ‘alleviates robot data scarcity’ via zero-shot latent, yet evaluation uses 200 trajectories per task for the Action Expert. This contradicts the zero-shot implication. The claim overstates generalization without the fine-tuning data.
- [science] Attributing performance gains solely to the ‘world latent’ is unsupported. Baselines use the same Action Expert and training data. The gap may be due to initialization or optimization, not the paradigm. Rule out confounders before claiming causal efficacy.
- [writing] Listing ‘Quantum’ and ‘Proteins’ as future readout targets is a severe overreach for a 4B video-language model. No evidence or theory supports generalizing macroscopic video dynamics to quantum states. Remove or heavily qualify this speculative claim.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a novel approach to world modeling but lacks sufficient detail regarding safety protocols and ethical oversight for its physical experiments.

Ethical Oversight and Physical Safety: The paper details real-robot experiments involving dual-arm manipulation (Section 4.3, Appendix) where the model is explicitly tested on “failure” scenarios and “recovery” (Figures 5-9). While the tasks (e.g., stacking bowls, scooping sugar) appear benign, the use of physical hardware necessitates a clear statement regarding safety protocols. Specifically, the authors must confirm whether these experiments were conducted under an Institutional Review Board (IRB) or Institutional Animal Care and Use Committee (IACUC) equivalent for robotics, or if a specific safety review board approved the risk of physical damage to the robot or environment during the OOD (Out-of-Distribution) failure tests. The absence of such a statement in the "Conclusion" or "Appendix" is a significant omission for a paper involving embodied AI.

Automated Evaluation Safety: The evaluation of image and action generation relies heavily on LLM-based judges (Gemini 3.1 Pro, GPT 5.4, Doubao Seed 2.0 Pro) as described in Section 4.2 and the Appendix. The paper does not address the safety alignment of these judges. If the model generates a physically dangerous action (e.g., a robot arm moving towards a human, though not explicitly in the current tasks), an unaligned LLM judge might score it highly based on "instruction following" rather than safety. The authors should clarify if a safety filter or human verification step was applied to the outputs of these judges to prevent the propagation of unsafe behaviors in the evaluation metrics.

Data Privacy and Consent: The paper mentions using "12.5K h" of video data and "160M events" for pre-training (Appendix, Section 5.1). While the source is not explicitly detailed in the provided text, if this data includes human subjects (e.g., from AgiBot-World or similar datasets), the authors must explicitly state that the data was collected with appropriate consent and that privacy measures (such as face blurring or PII removal) were applied. The current text assumes the data is safe without justification.

Recommendation: The authors should add a dedicated "Ethical Considerations" or "Safety Statement" section. This section must:

1. Confirm IRB/ethics approval for real-robot testing.
2. Describe safety protocols used during physical experiments (e.g., emergency stops, human supervision).
3. Discuss the limitations of LLM judges in assessing safety and any mitigations used.
4. Clarify the provenance and consent status of the pre-training video data.

Action items.

- [science] The paper lacks explicit IRB/IACUC approval statements or ethical clearance for the real-robot experiments described in Section 4.3 (Appendix). Given the use of physical hardware and potential for physical harm during OOD failures, a formal safety protocol or ethics board approval statement is required.
- [science] The evaluation relies on LLM judges (Gemini 3.1 Pro, GPT 5.4) for scoring image and action generation (Section 4.2, Appendix). The paper does not address the potential for bias, hallucination, or safety misalignment in these automated judges, nor does it provide a human-in-the-loop verification step for safety-critical failures.
- [writing] The 'PRICE-V0.1' benchmark and real-robot tasks involve physical interactions. The paper does not explicitly discuss the safety measures taken to prevent physical damage to the robot or the environment during the 'failure' scenarios described in Figures 5-9 (Appendix). A safety mitigation statement is needed.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence supporting the central claim—that a unified "world latent" learned via next-state prediction drives superior performance across text, image, and action modalities—is currently insufficient to support the paper's conclusions without significant revision.

First, the causal link between the proposed "world latent" and downstream action capabilities is not rigorously established. In Section 1.2, the authors claim that action generation emerges from video pre-training without action labels. However, the ablation study in Table 4 (Section 3.4) is inconclusive regarding the specific contribution of the latent representation versus the base VLM (Qwen3.5). The baseline Qwen3.5-4B (which shares the same backbone) performs poorly on action tasks (12.4 average score), while Orca-4B achieves 36.6. While this suggests the pre-training objective matters, the paper fails to control for the possibility that the specific architecture of the "Action Expert" or the fine-tuning process (20k steps) is the primary driver, rather than the latent itself. A more rigorous ablation comparing Orca's latent against a randomly initialized latent or a latent from a standard VLM pre-trained on the same data is necessary to isolate the effect of the "world model" hypothesis.

Second, the statistical robustness of the real-robot evaluation is weak. The training data for the Action Experts is limited to 200 trajectories per task (Section 3.3, Appendix). With only five tasks and two OOD settings, the total number of evaluation episodes is small. The reported metrics (e.g., Success Rate, Drawdown Recovery Ratio) show improvements, but the paper does not provide confidence intervals, standard errors, or results of statistical significance tests (e.g., t-tests or bootstrap resampling). Given the high variance inherent in real-robot manipulation, claiming "superior recovery" based on point estimates from a small sample is scientifically premature.

Third, the evaluation of image prediction (PRICE-V0.1) relies exclusively on LLM judges (Gemini 3.1 Pro, GPT 5.4, etc.) without any validation against human ground truth. The paper asserts that these judges can reliably score "physical plausibility" and "scene consistency," but no inter-rater reliability analysis or correlation with human ratings is provided. If the LLM judges are biased or inconsistent, the quantitative results in Table 2 (showing Orca outperforming FLUX.2) lose their validity. The authors must either include human evaluation or demonstrate that their LLM judges correlate strongly with human assessments on a held-out validation set.

Finally, the scaling law analysis (Figure 1) lacks clarity on data quality. The x-axis represents "video hours," but it is unclear if this refers to unique content or includes duplicates. If the 12.5K hours contain significant redundancy, the observed loss reduction may not reflect true scaling behavior. The authors should clarify their data deduplication strategy and ideally plot loss against unique video duration to substantiate the claim of effective scaling.

In summary, while the results are promising, the evidence is currently too fragile to support the strong claims of a unified world latent driving multi-modal performance. The paper requires more rigorous statistical analysis, better-controlled ablation studies, and validation of automated evaluation metrics before it can be accepted.
Action items.

- [science] The claim that action generation emerges from video pre-training lacks causal evidence. The ablation study does not isolate the 'world latent' contribution from the base VLM's inherent capabilities. A controlled experiment is required.
- [science] The real-robot evaluation uses only 200 trajectories per task. With 5 tasks and 2 OOD settings, statistical power is low. Report confidence intervals or perform significance testing on FNS/DRR metrics to validate gains over pi_0.5.
- [science] The PRICE-V0.1 benchmark relies entirely on LLM judges without validation against human ground truth. Provide inter-rater reliability analysis or correlation with human ratings to support the quantitative claims of superiority.
- [science] The scaling law analysis does not clarify if the 12.5K video hours contain duplicates. Clarify the data deduplication strategy and plot loss vs. unique video duration to substantiate the scalability claim.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the evaluation section is insufficient to support the paper's central claims regarding the superiority of the Orca paradigm.

First, the ablation study (Section 5.3, Table 4) presents a critical statistical flaw. The authors report single-point average scores for Text, Image, and Action generation across different loss configurations. For complex generative tasks, especially those involving LLMs and robotics, performance is inherently stochastic. Reporting only the mean without standard deviations, confidence intervals, or results from multiple random seeds makes it impossible to determine if the observed improvements (e.g., the jump from 42.6 to 48.0 average) are statistically significant or merely noise. The claim that the joint objective is "most balanced" is qualitative; a quantitative statistical test (e.g., ANOVA or paired t-tests) is required to validate this.

Second, the real-robot evaluation (Section 5.2.3, Table 3) lacks essential experimental design details. The rule-based scores (e.g., Orca-4B achieving 36.6 vs. $\pi_{0.5}$'s 31.2) are presented as definitive metrics. However, the manuscript does not specify the number of independent trials (*N*) conducted for each task and model configuration. Without *N*, the calculation of standard error is impossible. Furthermore, no statistical significance tests (such as a Mann-Whitney U test, which is appropriate for ordinal/ranked scores) are reported to confirm that the difference between Orca and the baselines is not due to chance. In robotics, where execution variance is high, a difference of ~5 points on a 100-point scale may not be significant without rigorous testing.

Third, the image prediction benchmark (Section 5.2.2, Table 2) relies entirely on LLM judges (Gemini 3.1 Pro, GPT 5.4, etc.) for scoring. While the authors report mean and standard deviation, they fail to address inter-rater reliability. When using multiple LLMs as judges, it is crucial to demonstrate that these judges agree with each other (e.g., via Cohen's Kappa or Kendall's W). If the judges are inconsistent, the reported "average" score is statistically meaningless. Additionally, the use of proprietary models as the ground truth for a scientific benchmark introduces potential bias that is not statistically controlled for.

Finally, the scaling laws (Section 5.1, Figures 1 & 2) are presented as smooth curves. While the trend is visually apparent, the paper does not provide the underlying data points, error bars for the loss values, or a fitted model with confidence intervals. This makes it difficult to assess the robustness of the claimed "scalability."

To proceed, the authors must re-run evaluations with sufficient trials to calculate confidence intervals, perform statistical significance testing on all comparative results, and report inter-rater reliability for LLM-based metrics.
Action items.

- [science] The ablation study (Table 4) reports single-point averages for complex downstream tasks (Text, Image, Action) without standard deviations, confidence intervals, or significance testing. Given the high variance in LLM benchmarks, claim 'All three objectives provide the most balanced downstream readouts' is statistically unsupported without error bars or p-values.
- [science] The real-robot evaluation (Table 3) uses a rule-based scoring system with integer points. The paper reports mean scores (e.g., 36.6 vs 31.2) but omits the number of independent trials (N) per task and model, and lacks statistical tests (e.g., t-test, Mann-Whitney U) to confirm if the observed differences are significant or due to random variance in robot execution.
- [science] The image prediction benchmark (Table 2) relies on LLM judges (Gemini, GPT-5.4) scoring on a 1-5 scale. The paper reports mean and standard deviation but does not address inter-rater reliability (e.g., Cohen's Kappa) or the potential bias of using proprietary models as ground truth, which undermines the statistical validity of the '59.8' score claim.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling technical contribution, but the writing quality is currently compromised by significant structural redundancy and minor inconsistencies that hinder readability.

The most critical issue is the duplication of content between the main evaluation sections and the appendix. Specifically, the content in `e001` (Evaluation) and `e002` (Loss of Proposed Learning Paradigm) is nearly identical, repeating the same scaling law figures, tables, and textual analysis. This suggests a copy-paste error or a failure to merge draft versions. The authors must consolidate these sections into a single, logical flow. The current structure forces the reader to encounter the same data and arguments twice, which disrupts the narrative cohesion.

Additionally, there are minor inconsistencies in terminology. For instance, the Appendix lists "SmolVLM2" as a baseline, while the main text and bibliography refer to "SmolVLM". Such discrepancies, while small, reduce the professional polish of the paper. Furthermore, several figure captions in the provided source text appear truncated (e.g., ending abruptly with "Unconsciou..."), indicating incomplete editing.

The prose itself is generally clear and concise where it appears, but the structural issues described above must be resolved to meet the standards of a high-quality publication. The authors should perform a thorough pass to ensure all sections are unique, all references are consistent, and all captions are complete.

Action items.

- [writing] The manuscript contains significant structural redundancy. Sections 'Evaluation' (e001) and 'Loss of Proposed Learning Paradigm' (e002) repeat identical content, figures, and tables. The authors must consolidate these into a single, coherent narrative flow to improve readability and remove confusion.
- [writing] In Section 'Evaluation Settings' (Appendix), the list of baselines includes 'SmolVLM2', but the bibliography and main text consistently refer to 'SmolVLM'. Ensure consistent naming conventions throughout the manuscript to avoid reader confusion.
- [writing] Several figure captions (e.g., Fig. 1 in e002) are truncated or incomplete in the source text (e.g., 'Unconsciou...'). Ensure all captions are fully written and grammatically complete before final compilation.